

LECTURE-6

Monday.
7/9/28

Count Inversions:

Problem:

Ans:

1	2	3	4	5
5	3	1	2	4

If $i < j$ and $A[i] > A[j]$; we increment a counter.
(where i is smaller than j are indexes)

counter = 0

for $i = 1$ to $n-1$

for $j = i+1$ to n

if $A[i] > A[j]$

count += 1

$i=1$

$j = 2, 3, 4 \dots n$ ($n-1$ times)

$i=2$

$j = 3, 4, 5 \dots n$ ($n-2$ times)

$i=3$

$j = 4, 5, 6 \dots n$ ($n-3$ times)

* For each 'i' iteration; inner loop runs $(n-i)$ times

$$T(n) = (n-1) + (n-2) + (n-3) \dots + 1$$

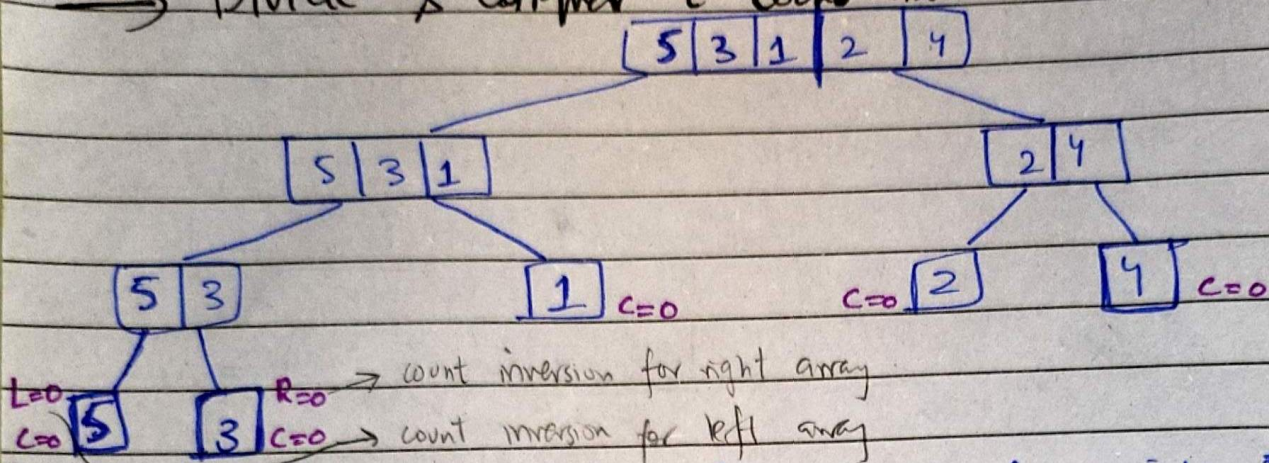
$$\therefore \text{Sum of Arithmetic mean} = \frac{n(n-1)}{2}$$

Time complexity = $O(n^2)$

Pearl Notes

Break the array to single elements each so the count inversion is zero as we don't have two indexes to compare (Date)

⇒ Divide & Conquer to "Count Inversions":



• Since [5] & [3] have $c=0$ (as leaf nodes & no index for comparison)

∴ we'll return to above level. But now while returning; we'll be using merge routine and return the sorted array to above level.

This will be done by $\begin{matrix} L & [5] \\ R & [3] \end{matrix}$ method where we'll compare L & R arrays & copy the lower data (i.e. 3). ALSO while copying from R array (if element of R is lower obviously); we'll always increment 'C' which is the count inversions.

* The Number of times to increment 'C' will be the no. of indexes remaining in the 'L' array (including current index) & this is because L array is sorted which means ^{all} the elements on right side of current index of L array must be higher in value IF & ONLY IF we're copying element from R array.

e.g: $\begin{matrix} L & [1] & [3] & [5] \\ R & [2] & [4] & \end{matrix}$ Since '2' copied for sorting L & R by merge routine (after 1); ∴ count = +2; as we are comparing 2 with '3' for copying & we've one more element (5) after 3.

Date _____

Max-Sub Array Sum:

6	-8	3	1	-5	9
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Search a mini sub array and

return the size of that sub array (i.e.: n).

Here the sum of that mini array's elements should be maximum. Sub array should've consecutive index. ABOVE answer $\Rightarrow$ max value is '9' & n=1

6	8	-3	11	-5	9
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$\rightarrow$ answer: n=4 i=1 $\rightarrow$ 4